StormCare RAG — Milestone 2 Implementation Report

(Track C: Retrieval System / RAG Foundation)
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1. Executive Summary

This milestone presents the foundational implementation of a **Retrieval-Augmented Generation (RAG)** system. The main goal is not to achieve high model performance yet, but to demonstrate that the **entire pipeline works correctly, end-to-end, and is reproducible**.

The system includes:

- A trained tokenizer with saved configurations and diagnostics
- A basic retrieval system using BM25
- A working training pipeline with a small Transformer model
- Logged training and validation losses to verify learning

The pipeline follows a clear flow:

PDF Documents → JSONL Corpus → Tokenization → Batching → Model Training → Loss Logging

This confirms that all components are properly connected and functioning.

2. Data and Preprocessing

The dataset is constructed from two hurricane-related PDF documents, chosen specifically for their relevance to storm analysis and the presence of rich domain-specific vocabulary. These PDFs are processed using the *pypdf* library, where the extracted text is divided into paragraph-level chunks to maintain meaningful context. During preprocessing, very short or irrelevant fragments are filtered out to ensure data quality. Each resulting chunk is enriched with metadata, including the source file name, page number, and a unique chunk ID for traceability. The processed data is stored in a JSONL format, where each line represents a single paragraph containing the text along with its associated metadata fields. To ensure robustness, the system includes fallback support, meaning that if the main dataset is unavailable, a smaller backup dataset is automatically used so that the pipeline remains functional. Additionally, reproducibility is maintained by using a fixed random seed (42) and predefined splits for training, validation, and testing, ensuring consistent results across runs.

3. Tokenizer and Vocabulary

A tokenizer is trained from scratch using the Byte Pair Encoding (BPE) method to effectively process the text data. The configuration includes a vocabulary size of 256, with normalization

applied through lowercase conversion and NFKC formatting, and pre-tokenization handled using whitespace splitting. Special tokens such as [PAD], [UNK], [BOS], and [EOS] are included to support sequence processing and model training. A smaller vocabulary size is intentionally chosen to enable faster training on CPU, reduce memory usage, and ensure efficient testing of the pipeline's correctness. In terms of performance, the tokenizer was evaluated on 91 documents, with an average of approximately 605 tokens per document and an unknown token rate of 0%, which is ideal. These results indicate that the tokenizer successfully captures the domain-specific vocabulary and represents the text data effectively.

4. Retrieval Component (RAG Foundation)

A **BM25-based retriever** is implemented to simulate RAG-style retrieval.

How It Works

- User query → ranked document chunks
- Returns top-k relevant passages
- Includes metadata for traceability

Example

Query: *"flooding storm surge"*
→ Retrieves relevant paragraph from hurricane report with high score

This confirms the retrieval system is functional and meaningful.

5. Model and Training Pipeline

5.1 Data Pipeline

The data pipeline begins by tokenizing the JSONL corpus, after which the tokens are combined into continuous sequences and divided into fixed-length blocks for training. The configuration uses a context length of 64 and a batch size of 8, with the primary task being next-token prediction.

5.2 Model Architecture

A small Transformer-based model called MiniGPT is used for this implementation. It consists of 2 layers, 4 attention heads, and an embedding size of 128, with a total of approximately 470,000 parameters. The model utilizes token embeddings, positional embeddings, and causal masking to ensure proper sequence learning and prevent access to future tokens during training.

5.3 Training Setup

The training process uses cross-entropy as the loss function and the AdamW optimizer with a learning rate of 3e-4. The model is trained for 80 steps, with gradient clipping applied to

maintain stability. The system is designed to run efficiently on CPU, while also supporting GPU execution if available.

6. Results and Evidence

The system successfully trains without errors.

Loss Behavior

- Initial loss: **~5.72**
- Final loss: **~4.96**

This steady decrease shows:

- Model is learning
- Data pipeline is correct
- Training loop is stable

A loss plot is also generated for visualization.

7. Limitations

Current limitations include:

- Only **BM25 retrieval** (no semantic/dense retrieval yet)
- No generator model using retrieved context
- No OCR support (only text-based PDFs)
- Small tokenizer and model size (for testing only)

8. Future Work (Next Milestone Plan)

Future Work and Planned Improvements

The next phase of this project will focus on transforming the current implementation into a fully functional Retrieval-Augmented Generation (RAG) system. This will involve upgrading the retrieval component by incorporating dense, embedding-based methods to improve semantic search capabilities. Additionally, a generator model (LLM) will be integrated to produce responses conditioned on the retrieved information. The system will also be enhanced by increasing the vocabulary size and overall model capacity to improve performance and scalability. Evaluation methods will be expanded to include more advanced metrics such as Recall@k and faithfulness measures, ensuring both retrieval quality and output reliability. Furthermore, ablation studies will be conducted to systematically analyze the impact of different components and design choices within the system.

9. Reproducibility

The system is designed to be fully reproducible using the provided scripts, ensuring that all steps of the pipeline can be consistently executed from a clean environment. The workflow includes key commands for converting PDFs into a JSONL corpus, training the tokenizer, running diagnostic checks, executing the model training process, and performing the retrieval demo. To maintain consistency across different runs and environments, all required dependencies are clearly specified and pinned in the requirements.txt file, allowing users to recreate the exact setup without compatibility issues.

10. AI Assistance Disclosure

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AI tools- GPT-5.2, were used to support certain aspects of the project such as debugging code, improving the overall structure, and refining the language for clarity and presentation. However, all key components of the work—including design decisions, data selection, interpretations, and the final code structure—were developed independently, ensuring that the core ideas and implementation remain original.

Final Summary (Simple One-Line Insight)

This milestone proves that a complete RAG pipeline—from raw PDFs to model training—works correctly, reliably, and is ready for scaling into a full intelligent system.